

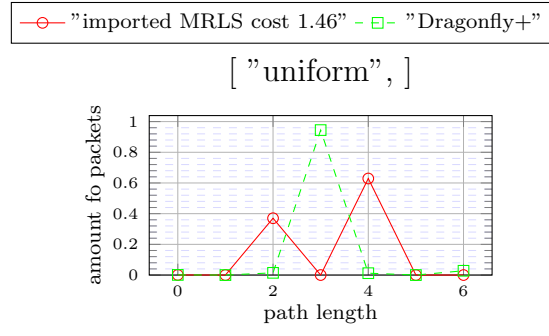


Figure 1: X: ["uniform",]

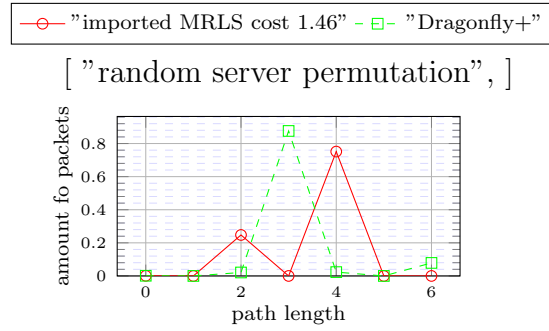


Figure 2: X: ["random server permutation",]

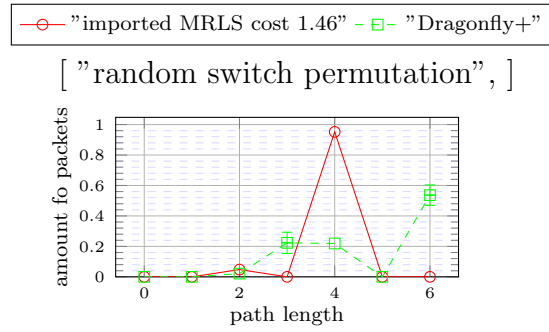


Figure 3: X: ["random switch permutation",]

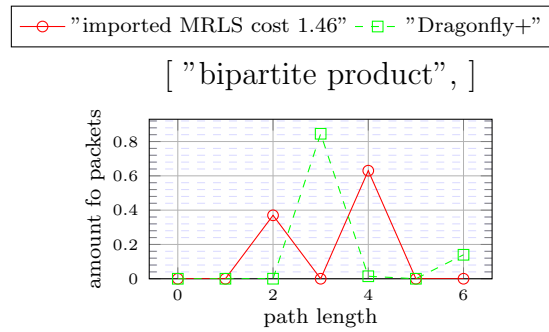


Figure 4: X: ["bipartite product",]

- heads/master-dirty-65e4f7db651d154c0abb3e3d0a691be7d6d92d82(0.6.3)

```
Configuration{
random_seed: ![ 42, 935, 128643 ],
warmup: 20000,
measured: 50000,
general_frequency_divisor: 2,
topology: topologia![
MultiStage{
stages: [
ExplicitStageFile{ filename: "..././MRLS_1280_19_760_32_a_net" }],
servers_per_leaf: 13,
legend_name: "imported MRLS cost 1.46"},
Megafly{ global_ports_per_spine: 16, servers_per_leaf: 16, global_arrangement: Palmtree, group_size: 16, number_of_groups: 65, legend_name:
"Dragonfly+" }],
traffic: HomogeneousTraffic{
pattern: ![
Uniform{ legend_name: "uniform" },
RandomPermutation{ legend_name: "random server permutation" },
CartesianTransform{
sides: topo![
[ 13, 1280 ],
[ 16, 1040 ]],
patterns: [ Identity, RandomPermutation ],
legend_name: "random switch permutation"},
Product{
block_size: 8320,
block_pattern: Uniform,
global_pattern: CartesianTransform{
sides: [ 2 ],
shift: [ 1 ]},
legend_name: ""},
legend_name: "bipartite product"}],
servers: 16640,
load: ![ 0.5 ],
message_size: 16},
maximum_packet_size: 16,
router: InputOutput{
virtual_channels: 4,
virtual_channel_policies: topologia![
[
MapHop{
hop_to_policy: [
ArgumentVC{
allowed: [ 0 ]},
ArgumentVC{
allowed: [ 0 ]},
ArgumentVC{
allowed: [ 1 ]},
ArgumentVC{
allowed: [ 1 ]},
ArgumentVC{
allowed: [ 2 ]},
ArgumentVC{
allowed: [ 2 ]},
ArgumentVC{
allowed: [ 3 ]},
ArgumentVC{
allowed: [ 3 ]}],
VecLabel{
label_vector: [ 0, 64, 80 ]},
OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
LowestLabel,
EnforceFlowControl,
Random],
[
MapHop{
hop_to_policy: [
MapLabel{
label_to_policy: [
Chain{
policies: [
OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0,
use_internal_space: true, use_neighbour_space: true, aggregate: false }]],
Chain{
policies: [
AverageOccupancyFunction{
label_coefficient: 0,
occupancy_coefficient: 2,
product_coefficient: 0,
constant_coefficient: 16,
use_internal_space: true,
use_neighbour_space: true,
exclude_minimal_ports: false,
virtual_channels: [ 0 ]}]}}],
MapLabel{
label_to_policy: [
Chain{
policies: [
OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0,
use_internal_space: true, use_neighbour_space: true, aggregate: false }]],
Chain{
policies: [
AverageOccupancyFunction{
label_coefficient: 0,
occupancy_coefficient: 3,
product_coefficient: 0,
constant_coefficient: 16,
use_internal_space: true,
use_neighbour_space: true,
```

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        exclude_minimal_ports: false,
        virtual_channels: [ 0 ],
        exclude_link_classes: [ 0 ]}}}}},
above_policy: MapLabel{
  label_to_policy: [
    Chain{
      policies: [
        OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space:
true, use_neighbour_space: true, aggregate: false }]],
      Chain{
        policies: [
          OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 48, use_internal_space:
true, use_neighbour_space: true, aggregate: false }]]}},
        LowestLabel,
        EnforceFlowControl,
        Random]],
allocator: Random,
buffer_size: 128,
bubble: false,
flit_size: 16,
intransit_priority: false,
allow_request_busy_port: true,
output_buffer_size: 64,
crossbar_frequency_divisor: 1,
crossbar_delay: 2},
routing: topologia![
  Polarized{ include_labels: true, strong: false, legend_name: "Polarized" },
  MegaflyAD{
    first_allowed_virtual_channels: [ 0, 1 ],
    second_allowed_virtual_channels: [ 2, 3 ],
    minimal_to_deroute: [ 1, 1, 0 ],
    source_group_pattern: [
      [ None, None ],
      [ None, None ]],
    intermediate_group_pattern: [
      [ None, None ],
      [ None, None ]],
    destination_group_pattern: [
      [ None, None ],
      [ None, None ]],
    global_pattern_per_hop: [
      [ None, None ],
      [ None, None ]],
    legend_name: "FPAR 2 vc"}],
link_classes: [
  LinkClass{ delay: 2 },
  LinkClass{ delay: 2 },
  LinkClass{ delay: 2 }],
launch_configurations: [
  Slurm{ job_pack_size: 1, time: "3-00:00:00" }]]}

```